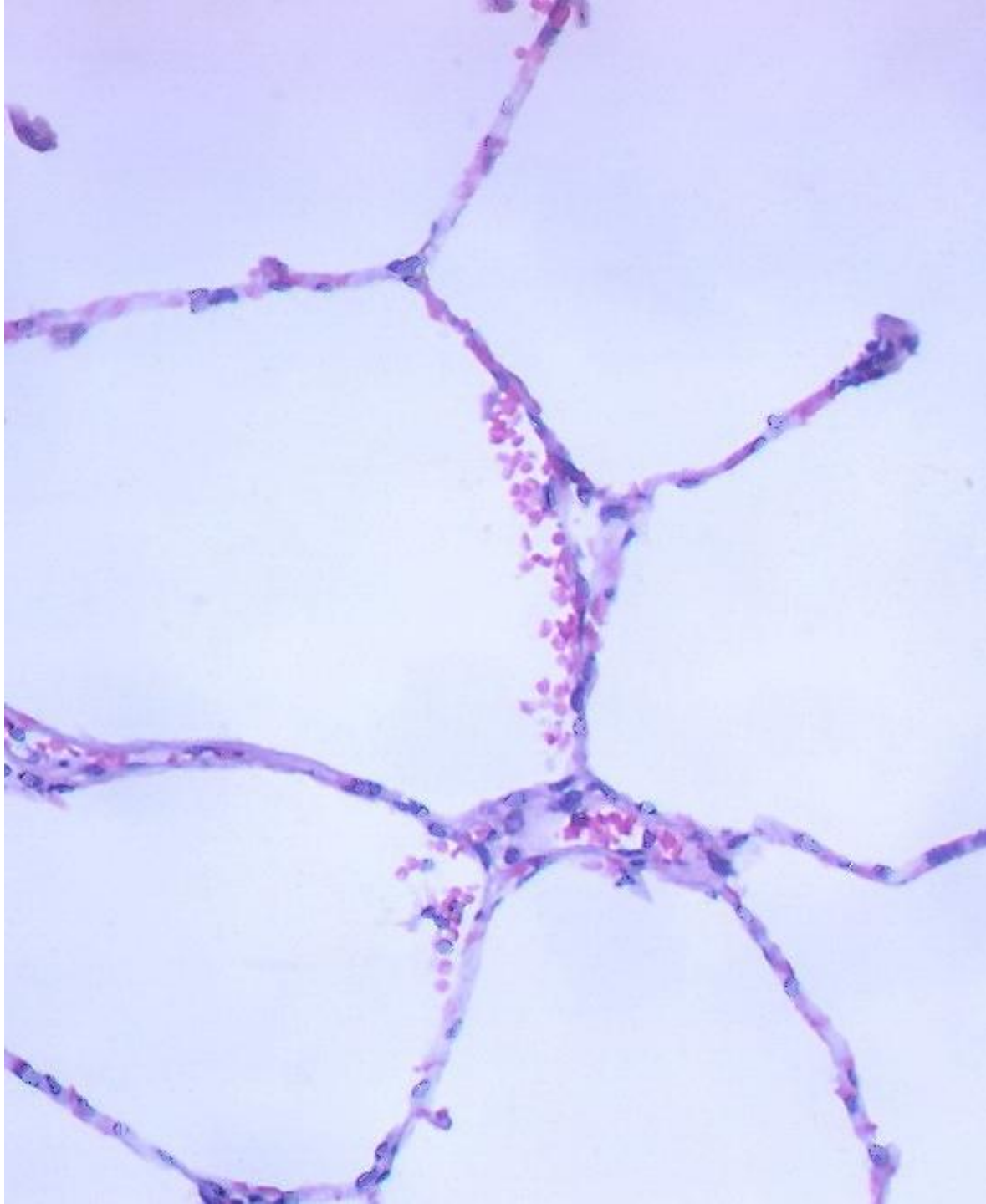
DP Biology – B3.1

Gas exchange

- What adaptations let multicellular organisms exchange gases?
- How do flowering plants and mammals differ, and resemble each other, in gas exchange?

Animal statements B3.1.1–6, SL and HL. Textbook pp. 273–283. Plants are in Plant Systems.

Image: human lung tissue, stained section. Yale Rosen, CC BY-SA 2.0.

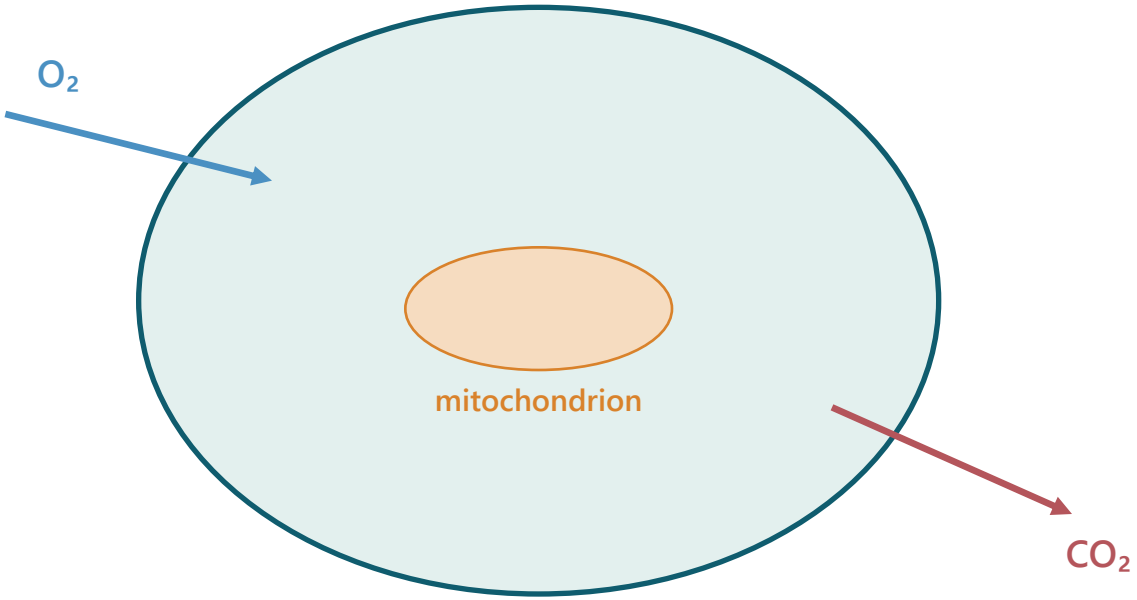


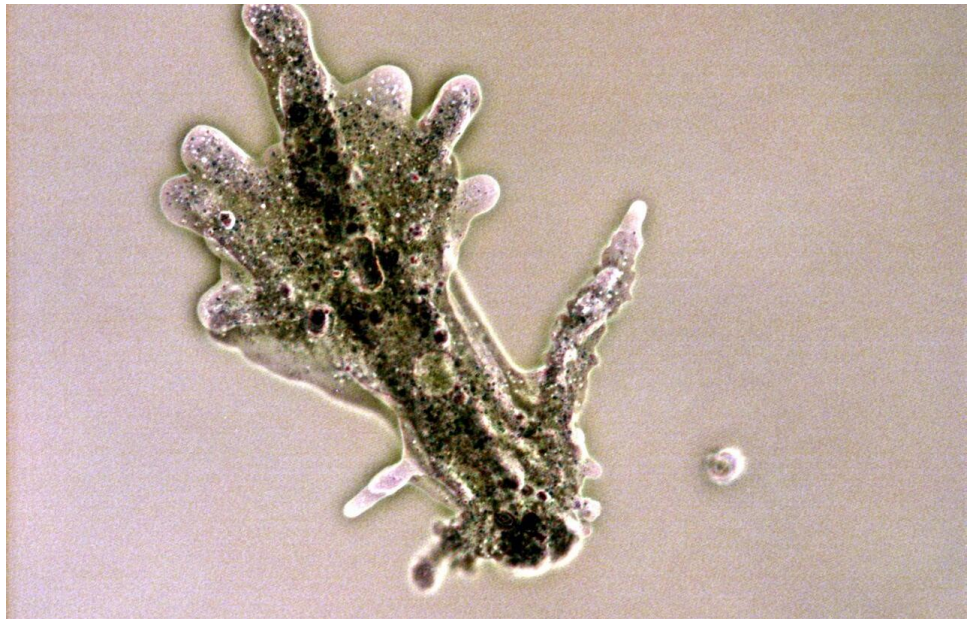
Objectives for each class

Class	Statements	By the end of the class, you can…
1	B3.1.1–3	Explain why size makes gas exchange harder, and how surfaces and gradients overcome this
2	B3.1.4	Explain how alveoli, bronchioles, capillary beds and surfactant suit the lungs to gas exchange
3	B3.1.5–6	Explain how muscles ventilate the lungs, and determine lung volumes from a spirometer trace

Every cell must exchange gases

B3.1.1



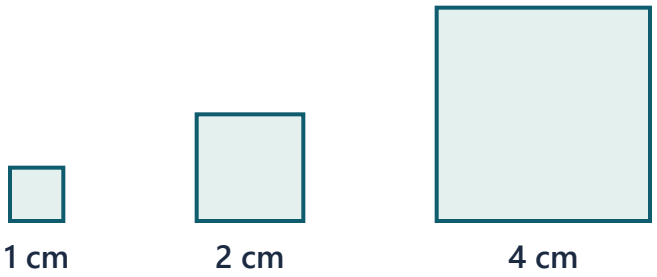


Amoeba proteus, a single cell. Cymothoa exigua, CC BY-SA 3.0.

Cells respire all the time. Aerobic respiration **uses oxygen** and **produces carbon dioxide**, and both gases cross membranes by **diffusion**.

In a single cell like Amoeba, the whole membrane is the exchange surface. That is enough because the cell is tiny.

Size makes gas exchange harder



Cube side	Area	Volume	Area : volume
1 cm	6 cm^2	1 cm^3	6 : 1
2 cm	24 cm^2	8 cm^3	3 : 1
4 cm	96 cm^2	64 cm^3	1.5 : 1

As an animal grows, its **surface area-to-volume ratio falls**: less surface for each unit of respiring tissue.

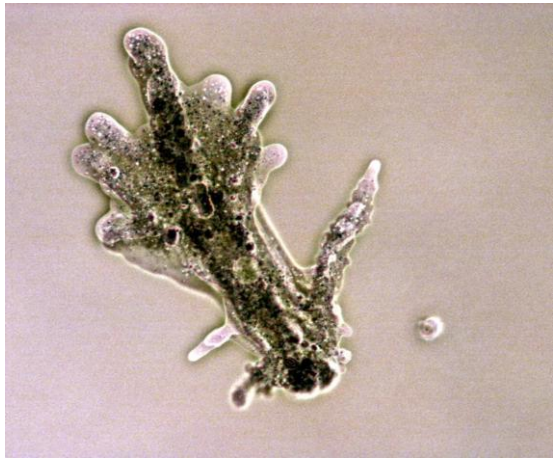
The **distance** from the centre to the outside also grows, and diffusion is only fast over tiny distances.

Key idea

Large, active animals need a specialized exchange organ and a transport system.

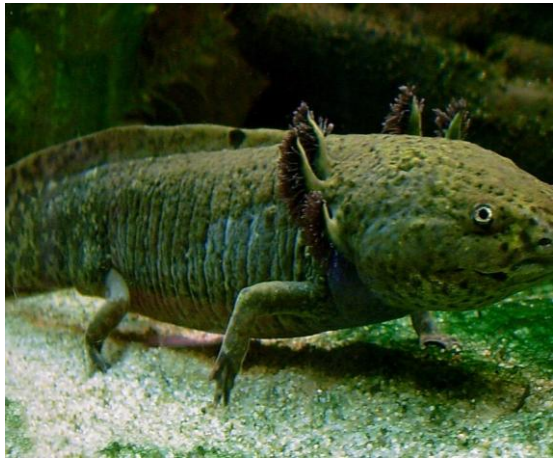
Different animals, different solutions

B3.1.1



Single cell

Across the membrane



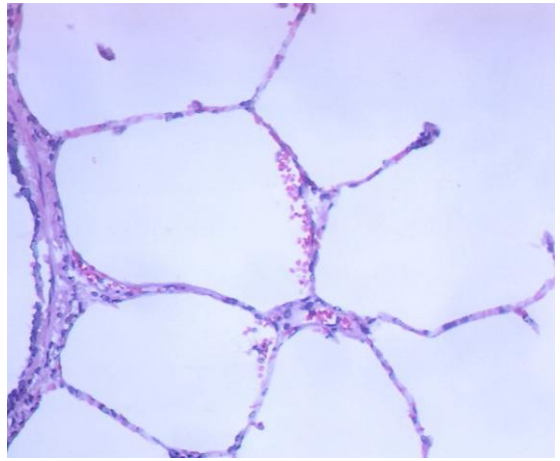
Salamander

External gills and skin



Bony fish

Internal gills



Mammal

Lungs with alveoli

Images, left to right: Cymothoa exigua, CC BY-SA 3.0; LoKiLeCh, CC BY-SA 3.0; Krishna satya 333, CC BY-SA 4.0; Yale Rosen, CC BY-SA 2.0.

Think

For each organism, explain how its size and habitat fit the way it exchanges gases. What would happen to the axolotl if its gills dried out?

Four properties of an exchange surface

B3.1.2

Permeable — Oxygen and carbon dioxide can pass through it.

Thin — The diffusion distance is as short as possible.

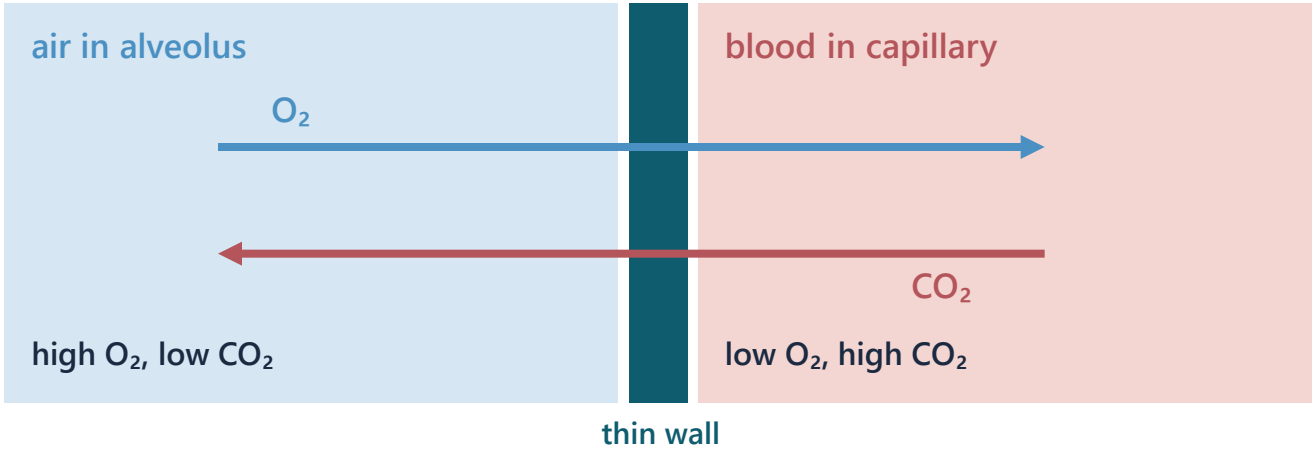
Moist — Gases dissolve first, which helps them cross.

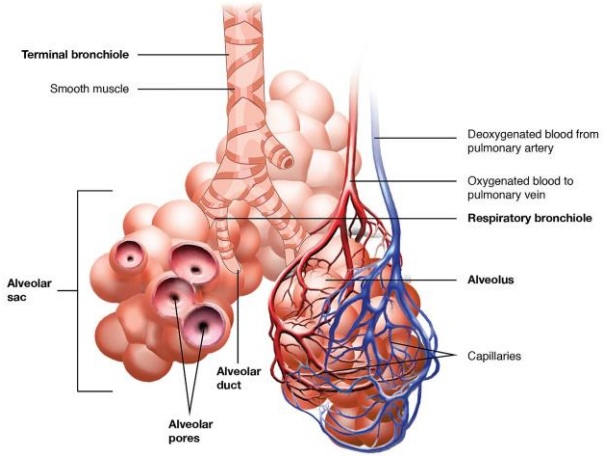
Large surface area — More gas can cross at the same time.

Link

The rate of diffusion depends on surface area, concentration gradient and distance. Every adaptation in this chapter changes one of these.

Keeping the gradient steep





Blood vessels around alveoli. OpenStax College, CC BY 3.0.

Dense capillary network

A large area of blood lies close to the surface.

Continuous blood flow

Takes oxygen away and brings carbon dioxide.

Ventilation

Replaces the used air or water with fresh.

Diffusion evens out concentrations. Without these three, the gradient would vanish and exchange would stop.

Gills do the same job in water

B3.1.3



Gills from a bony fish: many thin filaments with a dense blood supply. Krishna satya 333, CC BY-SA 4.0.

Thin **filaments** give a large area, and a dense network of **capillaries** runs through each one.

The fish pumps water over its gills: **ventilation with water**. Moving water brings oxygen and removes carbon dioxide.

Think

Water holds far less oxygen than air. Suggest why many fish die in warm, still water.

Practice: size and diffusion

B3.1.1

Cubes of pink agar were put in acid; the colour fades as acid diffuses in. Times are invented for practice; the pattern is realistic.

Cube side / cm	Area : volume	Time to clear / min
1	6 : 1	4
2	?	17
3	?	35

1. Calculate the area : volume ratio of the 2 cm and 3 cm cubes.
2. Describe the relationship between size and time.
3. Explain why a large animal cannot rely on diffusion through its skin.

Check-in: class 1

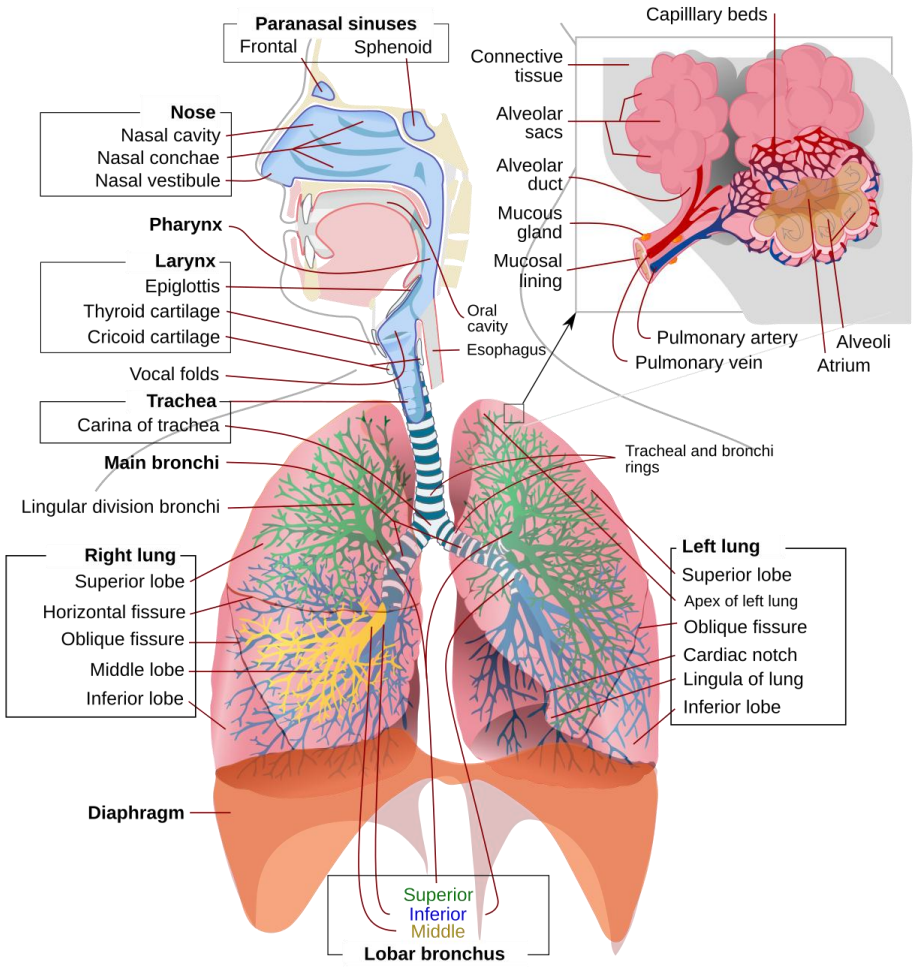
B3.1.1–3

1. State the two gases exchanged and the process that moves them.
2. Explain why size makes gas exchange harder.
3. State four properties of a gas-exchange surface.
4. Outline three ways concentration gradients are maintained.
5. Compare ventilation in a fish and a mammal.

How to use this

Answer without your notes, then check against slides 3–9.

From trachea to alveoli



Human respiratory system. LadyofHats, Jmarchn, public domain.

trachea → bronchi → bronchioles → alveoli

Rings of **cartilage** hold the trachea open. It splits into two bronchi, which divide again and again into a **branched network of bronchioles**.

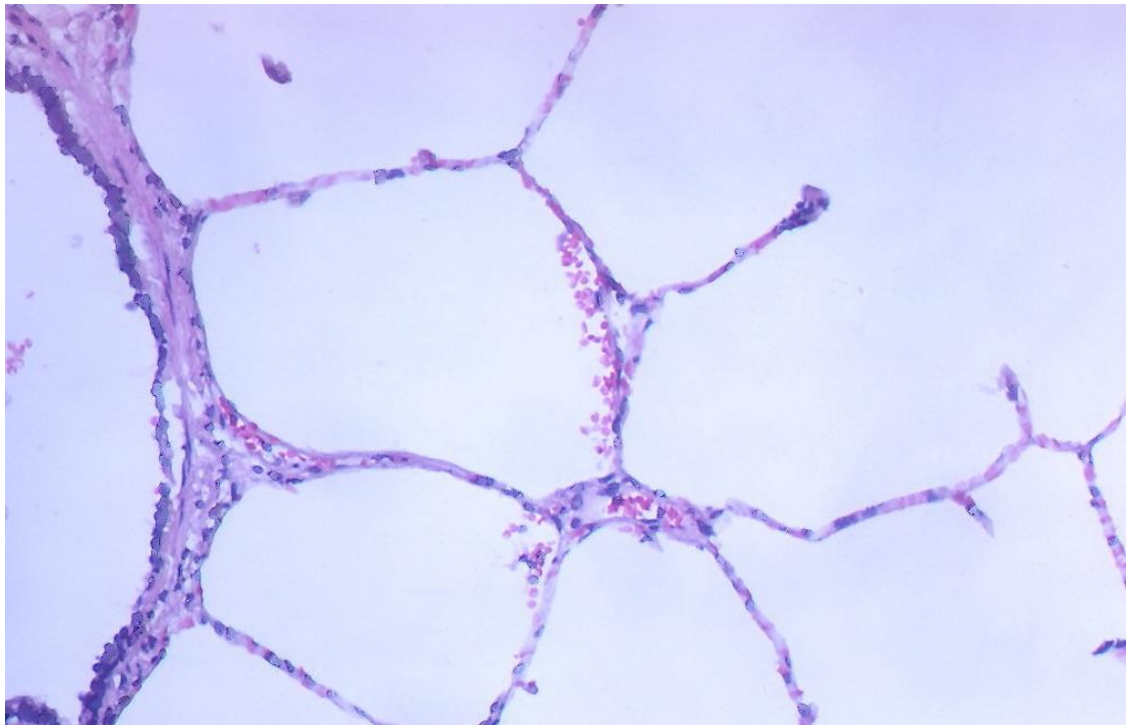
The finest bronchioles end in clusters of **alveoli**, where gas exchange happens. The branching spreads air to every part of the lung.

Note

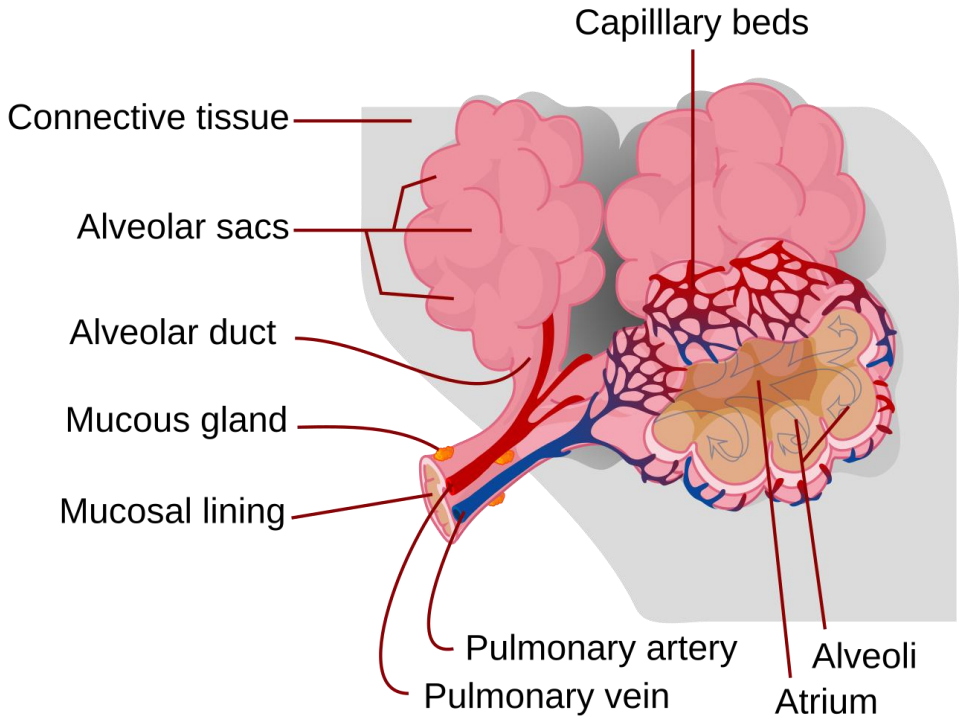
Air goes in and out by the same route, so some stale air always stays in the lungs.

Alveoli give a huge surface area

B3.1.4



Lung tissue: the spaces are alveoli, with very thin walls. Yale Rosen, CC BY-SA 2.0.

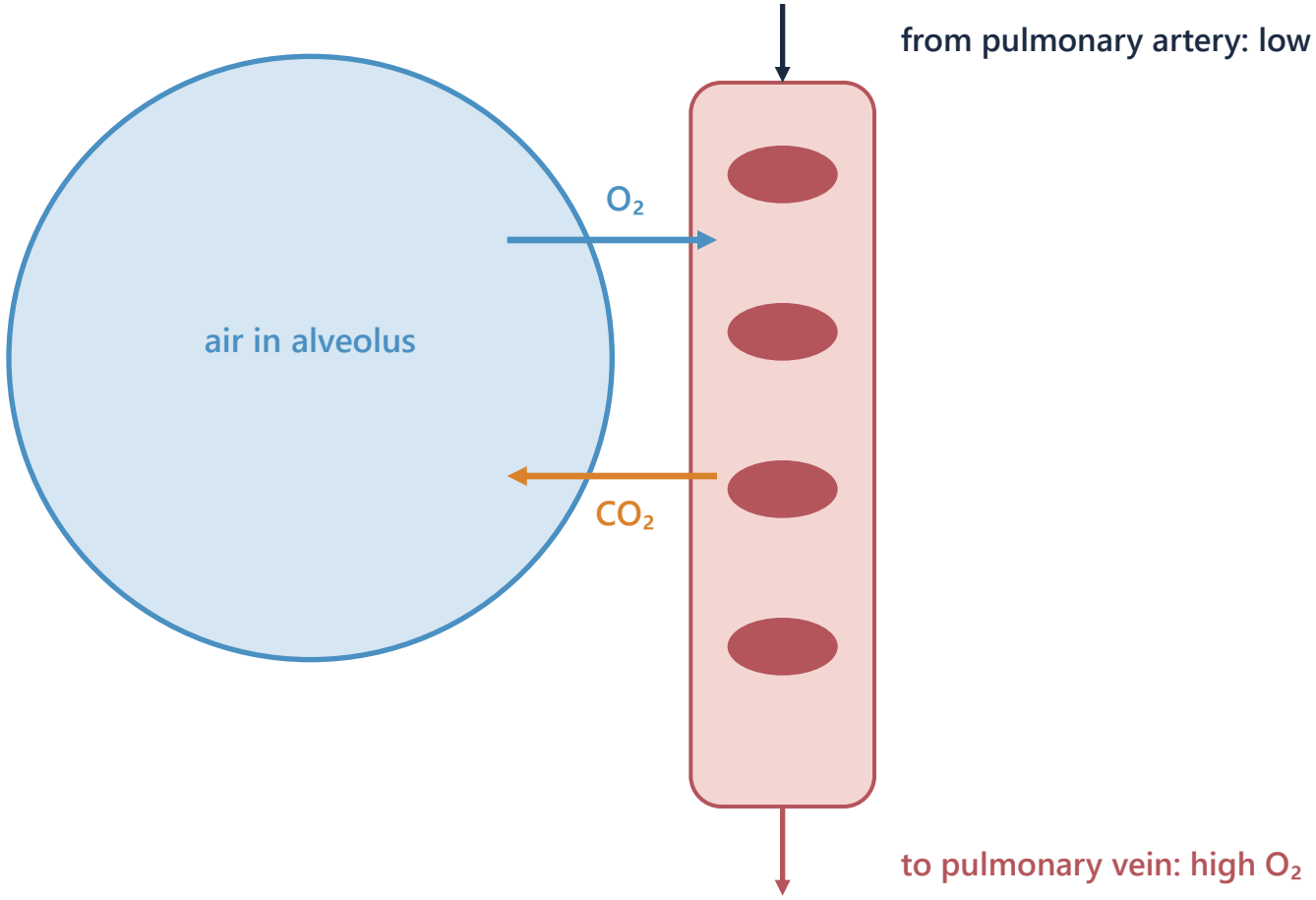


Alveoli and capillaries. LadyofHats, public domain.

Millions of tiny alveoli give an adult about **50 m²** of surface. The area comes from their **small size and huge number**, not their round shape.

Each **wall is one flattened cell thick**, and a moist lining lets oxygen dissolve.

Extensive capillary beds

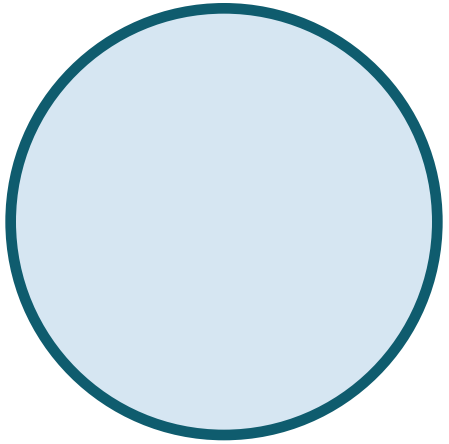


A dense **capillary bed** wraps every cluster of alveoli.

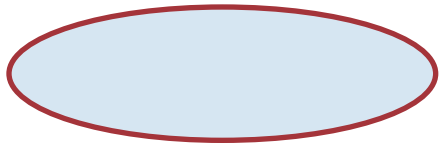
It adds to the surface area, and fast blood flow keeps the gradients **steep**.

Oxygen dissolves in the lining, diffuses across both walls and enters red blood cells.

Surfactant stops alveoli collapsing



with surfactant: stays open



without: walls stick together

Surfactant is a secretion of **phospholipids** from cells in the alveolar wall.

It **lowers the surface tension** of the water lining, so the moist walls do not stick together and the alveolus does not collapse when we breathe out.

Application

Foetal lungs make enough surfactant only late in pregnancy. Very premature babies can have stiff lungs and may be given surfactant.

Four adaptations of the lungs

B3.1.4

Adaptation	How it helps gas exchange
High surface area	Millions of tiny alveoli: more gas crosses at once
Branched bronchioles	Air reaches every alveolus in the lung
Extensive capillary beds	More area; blood flow keeps gradients steep
Surfactant	Lowers surface tension so alveoli stay open

Exam note

Learn these four by name. Then add thin walls and a moist lining from the four properties.

Working with data: damaged alveoli

B3.1.4

Emphysema destroys alveolar walls. Means for two groups of adults; the values are invented for practice, the pattern is realistic.

Measurement	Healthy	Emphysema
Alveolar surface / m^2	50	20
O_2 saturation at rest / %	97	91
O_2 saturation walking / %	96	83

1. Calculate the percentage fall in surface area.
2. Explain why blood oxygen is lower in emphysema.
3. Suggest why the difference is larger during exercise.

Check-in: class 2

B3.1.4

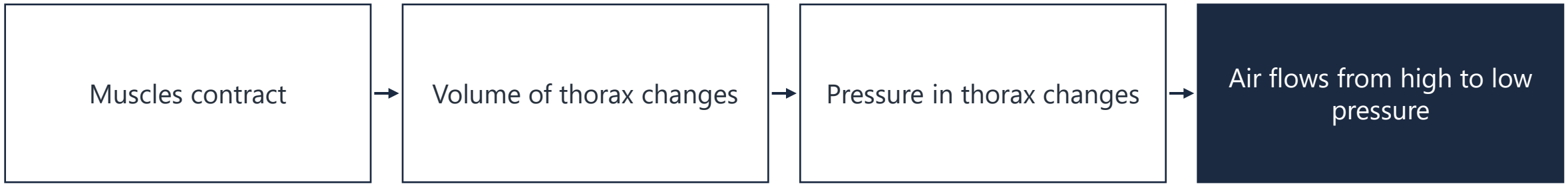
1. List the airways from trachea to alveoli.
2. Explain how the alveoli give a large surface area.
3. Describe the route of blood through a capillary bed in the lungs.
4. Explain the role of surfactant.
5. State four adaptations of the lungs for gas exchange.

How to use this

Answer without your notes, then check against slides 11–16.

Ventilation depends on pressure

B3.1.5



The thorax is an **airtight** chamber, so changing its volume changes the pressure inside. **Ventilation** keeps fresh air in the alveoli and the gradients steep.

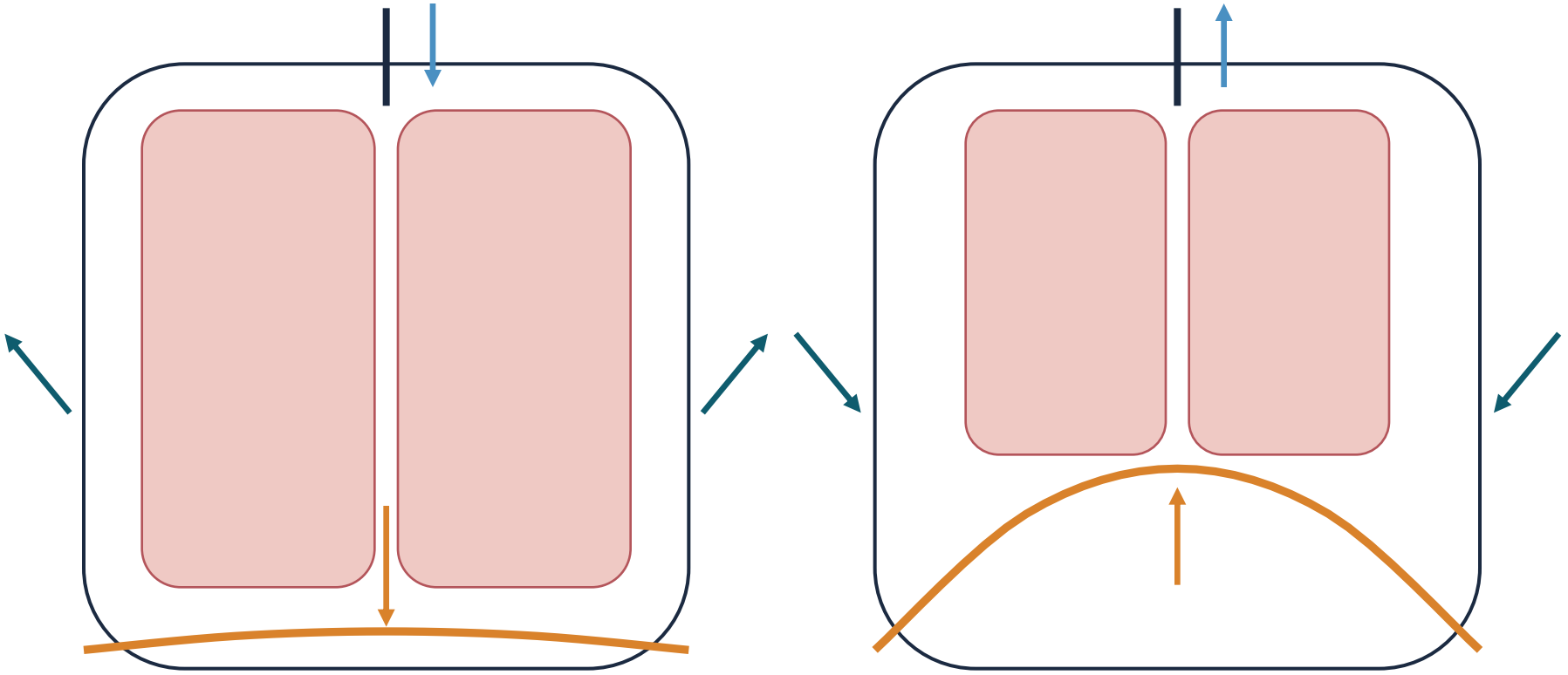
Muscles can only pull, so moving the ribs both ways needs an **antagonistic pair**.

Common mistake

Air coming in does not push the diaphragm down. The diaphragm moves down first, pressure falls, then air enters.

Breathing in and out

B3.1.5



Inhalation

Exhalation

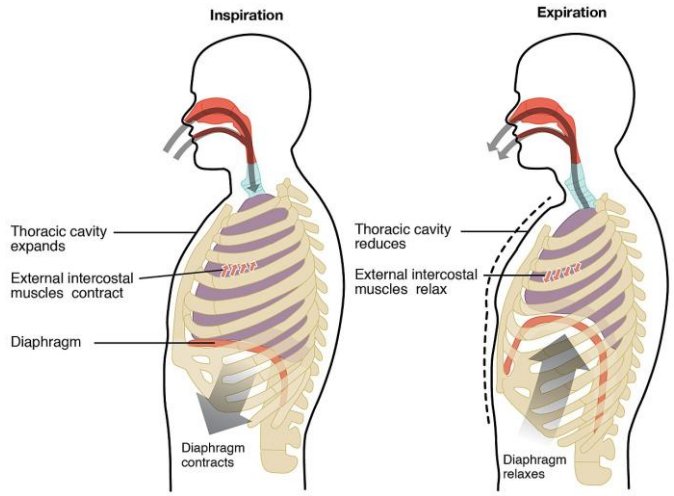
Drawn model. Orange: diaphragm. Teal arrows: ribcage. Blue arrow: air.

In: **diaphragm** contracts and flattens; ribs move **up and out**. Volume rises, pressure falls, air flows in.

Out: diaphragm relaxes into a dome; ribs move **down and in**. Volume falls, pressure rises, air flows out.

The muscles: inhalation and exhalation

	Inhalation	Exhalation
Diaphragm	Contracts, moves down	Relaxes, moves up
Abdominal muscles	Relax	Contract
External intercostals	Contract	Relax
Internal intercostals	Relax	Contract
Ribcage	Up and out	Down and in
Thorax volume	Increases	Decreases
Pressure	Below atmospheric	Above atmospheric
Air flow	In	Out



Inspiration and expiration. OpenStax College, CC BY 3.0.

How lung volumes are measured

B3.1.6



A portable spirometer. Biswarup Ganguly, CC BY 3.0.

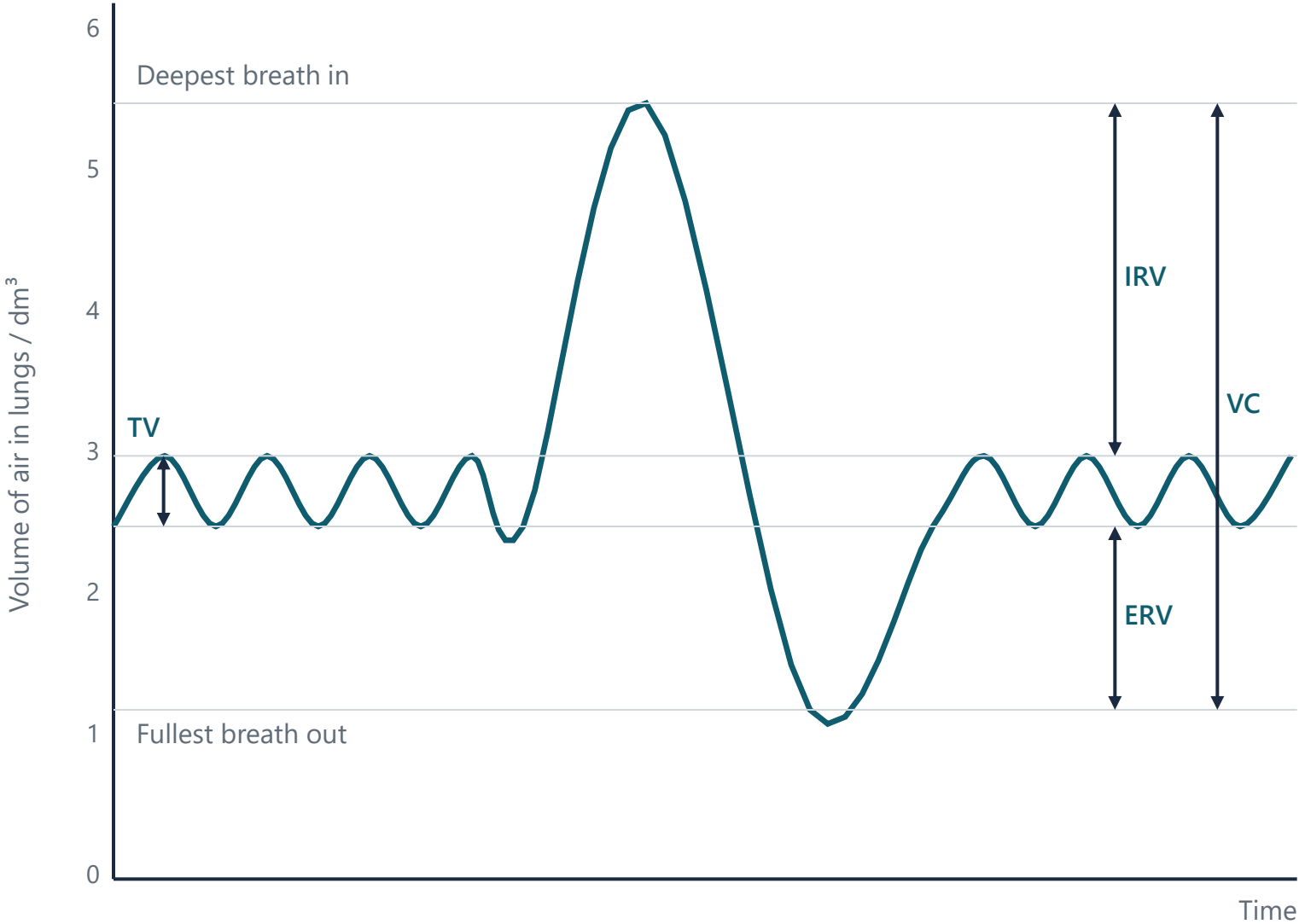
A **spirometer** records the volume of air breathed in and out over time.

Low-cost options: a lung volume bag, a balloon, or breathing out into an upturned bottle of water and measuring the water pushed out.

Why measure?

To compare rest and exercise, follow conditions such as asthma, or see the effect of training.

Reading a spirometer trace



Schematic trace with typical values.

- **Tidal volume**: one normal breath, about 0.5 dm^3
- **Inspiratory reserve**: extra air in after a normal breath in
- **Expiratory reserve**: extra air out after a normal breath out
- **Vital capacity**: deepest breath in to fullest breath out

Equation

VC = TV + IRV + ERV

Here: 0.5 + 2.5 + 1.3 = 4.3 dm^3

Practice: estimate your own lung volumes

B3.1.6

1. Breathe one normal breath into your own balloon; measure its diameter.
2. Volume of a sphere: $V = 4/3 \times 3.14 \times r^3$. Repeat three times; the mean estimates **tidal volume**.
3. Deepest breath in, then breathe out fully: this estimates **vital capacity**.
4. Count breaths in one minute: the **ventilation rate**.

Calculate

A balloon is 20 cm across after a full breath out. Estimate the vital capacity in dm^3.

Safety

Volunteers only. Never share a balloon or mouthpiece. Sit down if you feel dizzy.

Working with data: lung volumes and fitness

B3.1.6

Mean values for two groups of 17-year-olds, measured with a spirometer. These data are invented for practice; the pattern is realistic.

Volume / dm^3	Swimmers	Non-swimmers
Tidal volume at rest	0.5	0.5
Inspiratory reserve	3.3	2.8
Expiratory reserve	1.4	1.1
Vital capacity	5.2	4.4

1. Show how the swimmers' vital capacity was calculated.
2. Calculate the percentage difference in vital capacity.
3. Suggest why tidal volume at rest is the same.
4. Suggest a reason, other than training, for the difference.

Check-in: class 3

B3.1.5–6

1. Explain why changing thorax volume moves air.
2. Describe the muscle actions that cause inhalation.
3. State why two sets of intercostal muscles are needed.
4. Define tidal volume and vital capacity.
5. TV 0.5, IRV 3.0 and ERV 1.2 dm^3: calculate the vital capacity.

How to use this

Answer without your notes, then check against slides 18–24.

Common mistakes to avoid

Not this	But this
Alveoli have a large area because they are round.	The area comes from their small size and huge number.
Air rushing in pushes the diaphragm down.	The diaphragm moves down first; pressure falls; then air flows in.
Gas exchange happens in the bronchioles.	It happens in the alveoli; bronchioles carry air.
Breathing is the same as respiration.	Ventilation moves air; respiration is reactions in cells.
We breathe out only carbon dioxide.	Exhaled air still holds plenty of oxygen.

Summary: gas exchange in animals

B3.1.1–6

	Key idea
B3.1.1	Bigger animals have a lower area : volume ratio, so they need exchange organs
B3.1.2	Good surfaces are permeable, thin, moist and large
B3.1.3	Blood vessels, blood flow and ventilation keep gradients steep
B3.1.4	Alveoli, branched bronchioles, capillary beds and surfactant
B3.1.5	Muscles change thorax volume, then pressure, then air flows
B3.1.6	VC = TV + IRV + ERV, read from a spirometer trace

Check that you can...

1. **Explain** why size makes gas exchange harder
2. **State** four properties of a gas-exchange surface
3. **Outline** how concentration gradients are maintained
4. **Describe** the adaptations of the mammalian lung
5. **Explain** inhalation and exhalation in terms of muscles, volume and pressure
6. **Determine** lung volumes from a spirometer trace

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two properties of a gas-exchange surface, and how each one helps.

2. A student writes: "Air rushes in and pushes the diaphragm down." Correct this in one sentence.

3. Tidal volume is 0.5 dm^3, inspiratory reserve is 3.0 dm^3 and expiratory reserve is 1.0 dm^3. Calculate the vital capacity.

Image credits

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